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	Astronauts work in a dangerous and noisy environment, and have extraordinarily compressed work schedules; they are isolated from friends and family for long periods of time. Thus they may be fatigued, depressed, or anxious. Such stressors conspire to make individual performance, and that of the team as a whole, particularly vulnerable to cognitive decrements, and it is crucial to assess cognitive ability at regular intervals while astronauts work. In a project funded by the NSBRI, we have implemented a portable system, the MiniCog Rapid Assessment Battery (MRAB), which assesses 9 different cognitive functions. This system is intended to provide an "early warning alert," indicating when an astronaut or worker is suffering from stress-related deficits, such as fatigue, that may affect performance. One strength of our system is that the assessment is quick. The information can be used by the astronauts or workers themselves, either alone or as part of a team, or by a supervisor. In all cases, users can be warned to pay additional attention and take extra care, or better yet, take a break, consume food or caffeine, or even take a nap.		

Task Description:	In this phase of our research, we had proposed two major goals: First, we wished to extend the MiniCog system by a) programming the software to run on additional operating systems, b) allowing users to predict their performance before and after taking a test (but before seeing their scores, which would be compared with their predictions), c) adding wireless relay capability, and d) by including an additional initial screening test. Second, we planned validation studies of the MRAB tests, partly in collaboration with NSBRI team members David Dinges, Philip Lieberman, and Judith Orasanu. In our own lab, we had planned to conduct three kinds of studies: 1) correlating scores on MRAB with scores on assessment batteries currently used by NASA; 2) examining the effects on cognitive performance of four factors (anxiety, noise, circadian rhythm, and time-pressure); examining the mitigating effects of three potential countermeasures (a pre-learned short relaxation procedure, vigorous exercise, and caffeine). We now have a beta version of the MiniCog software for Windows PDAs. Both the Windows and Palm versions ask users to evaluate their own performance after they have completed a test, but prior to seeing their scores. The tasks have also been scripted for administration with commercial and free off-the-shelf experiment-presentation software such as E-Prime and PsyScope, and we implemented an online version of the tasks with beta software being developed for other projects in the lab. A completed study suggests that MRAB can detect selective effects of stress and sleep loss on specific cognitive abilities. Error rates (ERs) on the four MRAB tasks that are sensitive to variations in attention (vigilance, filtering, divided attention, and perceptual motor reaction time) were negatively correlated (in a range of -.36 to -.50, p<=.01) with the amount of sleep participants received the previous night. Furthermore, response times (RTs) on the same four tasks were positively correlated (in a range of .35 to .41, p<=.01) with participants' reported stress levels. A second study of stress and MRAB performance was initiated but relatively few participants were tested due to the lack of funds. Our initial testing protocol was not as successful at inducing high levels of anxiety as we had hoped, and we were not able to refine our protocol due to lack of funds. However, we were successful at leveraging our resources and introducing MRAB as an individual differences measure in a number of protocols. For instance, we discovered that the production of different types of lies (about the self versus another person or generated spontaneously versus from a rehearsed scenario) was predicted by distinct sets of MRAB scores, suggesting that deception is a complex process with the production of different kinds of lies depending on different cognitive processes. In a study of the cognitive styles of engineers versus humanities professionals, we administered the MRAB tests in conjunction with several other surveys and questionnaires, and found significantly different score profiles across the different professions. There has been sufficient interest in the task battery that we have distributed the software to a number of independent researchers, and we are collaborating with another lab to refine the battery for use in deception detection research.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The three general applications of this assessment toolbox have "earth" benefits as well as "space" benefits: First, the toolbox provides a practical means for assessing the effects of various stressors (such as fatigue, mood, noise, drug use, or diet) on cognitive processing. We not only expect some of those variables to have a significant impact on astronauts, but also expect them to affect people on earth (pilots, MDs, shiftworkers, and others). Second, the tasks can be used to assess the effectiveness of "countermeasures," such as training or drugs to manage fatigue, which may be as useful to workers on Earth as to astronauts in space. Finally, the tasks could be used by people "on the job" (e.g. truck drivers) to inform themselves about the current state of their cognitive processes. For example, when truck drivers "weigh in" after many hours of travel, they can run through the battery of tests and make sure that they are still mentally alert. Depending on their scores, they may want to rest before going back on the road, or they may eat a meal or drink coffee, or engage in other activities to reduce the danger of driving. Other researchers will be able to use our test battery to assess cognitive performance quickly and under a variety of laboratory and field conditions; they may also use the MiniCog platform to develop their own psychological tests or experiments. Both our test battery and the MiniCog application have an advantage over standard task batteries and many typical psychological scripting programs in that the tests are brief and the method of administration is compact, portable, and relatively inexpensive. This could make MiniCog and MRAB practical in a wide range of settings where there are questions of neurocognitive capability. The MRAB may also turn out to be a useful individual differences assessment tool, helpful in predicting performance on different types of professional tasks even in non-stressful situations.
Task Progress:	First, we have been successful in extending the MiniCog system; we now have a beta version of the MiniCog software for Windows PDAs. Both the Windows and Palm versions ask users to evaluate their own performance after they have completed a test, but prior to seeing their scores. The tasks have also been scripted for administration with commercial and free off-the-shelf experiment-presentation software such as E-Prime and PsyScope, and the software and task battery have been licensed to many other researchers. During the second year we initiated a study to address the following question: Is the MRAB capable of detecting the effects of induced anxiety on very specific aspects of performance? 31 participants came to the lab for 3 one-hour practice sessions within a 2-week period. During this time they practiced the tasks 8 times. At the end of the 2-wk period participants returned to the lab for a final session, during which they completed the MRAB tests and a series of mood and cognitive state questionnaires before and after receiving instructions about a stressful public-speaking task they would perform, and then again after they had performed this task. We had hoped with this task to simulate the stress induced by extremely high pressure work situations, and assess which, if any, cognitive functions would decline. Results: Mean state anxiety scores on the Spielberger State Anxiety Inventory (SA) changed significantly only in females, from 38 at the beginning of the session to 47 upon learning that there would be a public-speaking task (and 46 at the end of the session, which was not a significant decrease); in males the scores were 31 and 34 (and 33 at the end), respectively. In contrast, Spielberger reports increases to 50 and 61 (males and females) during a stressful movie. We were hoping our task would induce stress levels similar to the movie condition, but this did not occur, and with the funding shortage, there was no opportunity to redesign the protocol and administer it to more participants. In looking at the MRAB scores of just the female participants during the final session, we found an increase from baseline (at the

	beginning of the session) in ERs on the "cognitive set switching" task (from 1.85% to 4.63% [p<.01] upon learning about the public speaking task) and on the "filtering" task (from 2.91% to 5.49% [p<.01] at the end of the session). However, RTs decreased on other tasks: MR, SWM, and VWM (all tasks for which there were non-significant increases in ERs). Feasibility testing was conducted by including a subset of the MRAB tests in the NEEMO 9 mission in the spring of 2006, and we received useful feedback from the crewmembers. Leveraging resources, we have successfully added the MRAB tests as cognitive individual differences measures in a number of protocols for other research projects.
Bibliography Type:	Description: (Last Updated: 01/31/2008)
Articles in Peer-reviewed Journals	Shephard JM, Kho S, Chen J, Kosslyn SM. "MiniCog: a method for administering psychological tests and experiments on a handheld personal digital assistant." Behav Res Methods. 2006 Nov;38(4):648-55. PMID: 17393836 , Nov-2006